

Li Yuan (Leon)

Ph.D. Candidate in Statistical Science, Southern Methodist University

M.S. in Data Science, Vanderbilt University

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RESEARCH INTERESTS

Bayesian inference; nonlinear time series modeling and forecasting; interpretable machine learning; uncertainty quantification; statistical methods for text and biomedical data.

EDUCATION

Southern Methodist University, Dallas, TX

Expected May 2027

Ph.D. in Statistical Science

- Advisor: Raanju R. Sundararajan. Advanced to candidacy February 13, 2025. GPA: 3.87/4.00.
- Research: interpretable Bayesian models for nonlinear time series and text classification.

Southern Methodist University, Dallas, TX

December 2025

M.S. in Statistical Science GPA: 3.87/4.00.

Vanderbilt University, Nashville, TN

May 2022

M.S. in Data Science GPA: 3.89/4.00; 48 semester credits.

- Selected coursework: Data Management Systems; Modeling & Machine Learning I–II; Data Science Algorithms; Big Data Scaling; Social Network Analysis; Capstone Development.

Sichuan University, Chengdu, China

June 2019

B.S. in Statistics

- Completed two years of foundational mathematics coursework followed by two years of specialized training in statistics.

HONORS AND AWARDS

Scheuren Award, Southern Methodist University

May 2023

Outstanding performance on the theory component of the Ph.D. qualifying examination.

PEER-REVIEWED PUBLICATIONS

1. Chen, C., Ashana, D. C., Callison, K., Baker, C., Burns, R. N., Cao, J., **Yuan, L.**, and Gershengorn, H. B. (2026). Critical illness outcomes of hospitalized pregnant women following a Texas abortion ban. *American Journal of Respiratory and Critical Care Medicine*, 212(8), 1730–1739. [doi:10.1093/ajrccm/aamag207](https://doi.org/10.1093/ajrccm/aamag207).
2. Yan, B., Ran, X., Jiang, Y., Torrence, S. K., **Yuan, L.**, Shao, Q., and Yang, Z. J. (2021). Rate-perturbing single amino acid mutation for hydrolases: A statistical profiling. *The Journal of Physical Chemistry B*, 125(38), 10682–10691. [doi:10.1021/acs.jpcc.1c05901](https://doi.org/10.1021/acs.jpcc.1c05901).

MANUSCRIPTS UNDER REVIEW AT PEER-REVIEWED JOURNALS

1. **Li Yuan** and Raanju R. Sundararajan (2026). [BPAR: Interpretable Bayesian Bernstein Autoregression for Non-linear Time Series Forecasting](#). Under review at *Knowledge-Based Systems*.
 - Develops Bayesian Bernstein polynomial models for nonlinear time-series forecasting with interpretable lag effects and predictive uncertainty. Evaluation on 19 real-world and chaotic benchmark series shows competitive accuracy with a parsimonious model.
 - [Code and benchmarks](#).

2. **Li Yuan, Jing Cao, and Raanju R. Sundararajan** (2026). [BAMIC: Interpretable Sentiment Classification with Bayesian Uncertainty and Word-Level Contributions](#). Under review at *Information Processing & Management*.
 - Develops a partially Bayesian sentiment classifier that provides word-level contributions and predictive uncertainty using only document labels. Evaluation on Wine, SST-2, IMDB, and Yelp examines accuracy, calibration, and uncertainty-guided human review.
 - [Code and benchmarks](#).

MANUSCRIPTS IN PREPARATION

1. Rios-Melendez, S., **Yuan, L.**, Zhang, S., Mourya, R., Shivakumar, P., Alonso, E., Guthery, S., Harpavat, S., Horslen, S., Kamath, B., Karpen, S., Kohli, R., Loomes, K., Magee, J., Miethke, A., Shneider, B., Sokol, R., Valentino, P. L., Bezerra, J., Pandurangi, S., and the Childhood Liver Disease Research Network (ChiLDRN). Novel point-of-care diagnostic risk model and clinical calculator for biliary atresia. Manuscript in preparation for submission to *Hepatology Communications*, a peer-reviewed, open-access journal.
2. **Li Yuan, Raanju R. Sundararajan, and Chul Moon** (2026). Non-linear Lagged Dependence of ADHD fMRI Data. Manuscript in preparation.
 - Uses matrix auto-distance correlation to investigate nonlinear lagged dependence in ADHD-200 resting-state fMRI. Examines brain-network patterns and sensitivity to global signal regression through descriptive, hypothesis-generating analyses.

RESEARCH EXPERIENCE

Southern Methodist University

Dallas, TX

Doctoral Researcher, Statistics and Data Science

September 2024 – Present

- Develop interpretable Bayesian forecasting models and reproducible benchmark software with Raanju R. Sundararajan; investigate Bayesian attention models for sentiment classification with Sundararajan and Jing Cao.
- Investigate nonlinear lagged dependence in resting-state fMRI from ADHD-200 using matrix auto-distance correlation; ongoing methodological and computational work.

UT Southwestern Medical Center

Dallas, TX

Research Collaborator, with Sindhu Pandurangi

2024–2026

- Developed logistic regression models for biliary atresia risk using clinical variables and biomarkers, with internal and external validation analyses.
- Built an R Shiny research calculator to communicate individualized model predictions; co-author on the manuscript in preparation.

Southern Methodist University

Dallas, TX

Graduate Research Assistant, with Jing Cao and Catherine Chen (UTSW)

2024

- Contributed statistical analysis in R to a collaborative study of critical illness outcomes among hospitalized pregnant women, resulting in the *AJRCCM* publication listed above.

Southern Methodist University

Dallas, TX

Graduate Research Assistant, Statistics and Data Science

August 2023 – January 2024

- Applied topological data analysis to lung and brain cancer imaging and related research questions.

Data Science Institute, Vanderbilt University

Nashville, TN

Graduate Research Assistant, with Charreau Bell

August – December 2021

- Applied transformer models to classroom discourse classification and used Wav2Vec2 audio embeddings, dimension reduction, and clustering to analyze teacher speech with human-labeled comparison data.

Vanderbilt University School of Medicine
Part-Time Research Employee, with Regina Russell

Nashville, TN
June – October 2021

- Used R to analyze medical students' medical records and educational performance data collected during their studies at Vanderbilt. Generated individualized R Markdown reports to help students understand their progress and identify opportunities to improve their medical-school performance.

Vanderbilt University Law School
Graduate Research Assistant, with Tracey George, Albert Yoon, and Mitu Gulati

Nashville, TN
March – September 2021

- Converted and extracted structured fields from 3,365 acquisition-agreement documents using R; analyzed missing biographical information and gender patterns in legal transactions.

Vanderbilt University, Department of Chemistry
Graduate Research Assistant, with Zhongyue J. Yang

Nashville, TN
September 2020 – September 2021

- Integrated protein structure and enzyme kinetics data using Python, the Protein Data Bank, and GraphQL APIs; contributed to the hydrolase study published in *The Journal of Physical Chemistry B*.

TEACHING EXPERIENCE

Southern Methodist University
Graduate Teaching Assistant, STAT 2331: Introduction to Statistical Methods

Dallas, TX
Fall 2025

- Led three lab sessions each Friday, demonstrating basic data analysis in Excel and guiding students through lab work in a non-calculus introductory service course.
- Held office hours and provided individual guidance during lectures, in-class quizzes, and homework activities, helping students develop their reasoning without supplying answers.
- Graded all homework and labs, returned timely feedback, and proctored exams. Topics included descriptive statistics, regression, probability, and inference.

Southern Methodist University
Graduate Teaching Assistant, STAT 3300: Applied Statistics: Regression

Dallas, TX
2022–2023

- Graded all homework assignments and labs for an applied regression course covering statistical modeling, diagnostics, inference, and prediction.

Teaching interests: Introductory statistics; regression; computing with R and Python; data management; machine learning; probability and mathematical statistics; time series; Bayesian statistics; statistics and data science capstones.

STATISTICAL CONSULTING AND INDUSTRY EXPERIENCE

SMU Statistical Consulting Center
Statistical Consultant, Revision Skincare; supervised by Charles South

Dallas, TX
April – June 2025

- Analyzed repeated-measures skincare outcomes with paired nonparametric tests, multiplicity adjustment, and linear and robust mixed-effects models.
- Prepared a client presentation and reproducible Quarto analysis; also analyzed blinded facial-skin grading data across skin types and image views.

Deason Criminal Justice Reform Center, SMU
GIDEON Project, STAT 6366 Statistical Consulting; with Hunter Schuler

Dallas, TX
Fall 2023

- Linked case, event, and name tables into an analysis dataset of more than 390,000 rows. Constructed event-rate measures and applied count and time series models to public-defense data.
- Delivered an R Shiny dashboard and written report for client discussion.

- Developed time series models in R and Python to forecast daily Alexa performance indicators and identify anomalous observations, balancing false alarms against detection of genuine performance changes using prediction intervals.
- Compared Holt–Winters, seasonal ARIMA, dynamic regression with seasonal ARIMA errors, Prophet, Neural-Prophet, neural autoregressive models, and ensembles using time series cross-validation.
- Evaluated forecast accuracy and interval quality using MAE, RMSE, MAPE, coverage probability, and interval scores; developed a bootstrap-based forecasting ensemble with prediction intervals.

SELECTED COMPUTATIONAL PROJECTS

- **Medical image annotation (Vanderbilt M.S. capstone, 2022):** Developed a PyTorch workflow combining CLIP-generated pseudo-labels with a CNN for chest-versus-breast image classification on MedMNIST; compared held-out predictions across repeated random splits.
- **Amazon review analytics (Vanderbilt, DS 5460, team project, 2022):** Collaborated on processing 21.78 GB of reviews across 12 product categories using PySpark on Google Cloud/Hadoop; explored recommendation methods and fine-tuned a transformer sentiment model on sampled reviews.
- **U.S. traffic-accident database (Vanderbilt, DS 5420, team project, 2021):** Co-developed a MySQL-backed R Shiny application; contributed schema and functional-dependency checks, stored procedures and transactions, the database connection, and four interactive pages.
- **Formula One analysis (SMU, STAT 6341, 2023):** Fit a Bayesian mixed model in RStan with constructor and driver effects; compared predictive models and created a Shiny dashboard.
- **Geometric analysis of handwritten digits (SMU, STAT 6383, 2025):** Used elastic shape distances and Karcher means for classification; parallelized computations in R.

COMPUTING AND METHODS

- **Languages and tools:** R, Python, SQL, Excel, L^AT_EX, Git/GitHub, Quarto/R Markdown, Shiny, Jupyter.
- **Data systems:** MySQL, PySpark, Spark SQL/MLlib, Hadoop, Google Cloud.
- **Statistical computing:** RStan, tidyverse, tidymodels, lme4, robustlmm, PyTorch, transformers, scikit-learn, pandas, NumPy.
- **Methods:** Bayesian inference, variational inference, time series, regression and mixed models, statistical learning, uncertainty quantification, and model validation.
- **Languages:** Mandarin (native); English (fluent).

SELECTED ADDITIONAL TRAINING

- **SMU:** ESL 6001, Seminar for International Teaching Assistants, completed (Pass), Fall 2022.
- **Stanford University, online:** XCS224N, Natural Language Processing with Deep Learning (2021); CS234, Reinforcement Learning (2022).
- **Duke University, online:** Graduate-level Bayesian and Modern Statistics, Summer 2021.
- **University of Chicago, online:** Data & Policy Summer Scholar Program, 2021; capstone on retail mask policies, advised by Austin Wright.
- **University of California, Berkeley:** Global Access Program, August 2019–May 2020; Summer Session, June–August 2020.

REFERENCES

Raanju R. Sundararajan Dissertation advisor

Associate Professor, Department of Statistics and Data Science, Southern Methodist University

rsundararajan@smu.edu

Jing Cao Research collaborator

Professor and Chair, Department of Statistics and Data Science, Southern Methodist University

jcao@smu.edu

Charles South Consulting supervisor

Associate Professor of Practice; Director, Statistical Consulting Center, Southern Methodist University

csouth@smu.edu

Catherine Chen, M.D. Research collaborator

Associate Professor, Department of Internal Medicine

Division of Pulmonary and Critical Care Medicine, UT Southwestern Medical Center

catherine.chen@utsouthwestern.edu

Sindhu Pandurangi, M.D. Research collaborator

Assistant Professor, Department of Pediatrics, UT Southwestern Medical Center

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